

Gyro Logic Inference Rules

1. Observation Rules

R1 Query Generation: $(o \in O, S_t \in \Omega) \Rightarrow q_t^o = \text{GenQ}(o, S_t)$

R2 Slice Rule: $(q_t^o, E_o, S_t) \Rightarrow p_t^o = \text{Slice}(S_t, q_t^o, E_o)$

R3 Perception Non-Identity: $p_t^o \neq S_t$ (Perception is a slice of the world)

2. Evaluation Rules

R4 Truth Evaluation: $\tau_t^o = \text{Eval}_G(p_t^o, G_o, E_o)$

R5 Satisfaction: $\tau_t^o \geq \theta_o \Rightarrow (S_t, o) \models G \phi$

R6 Observer Relativity: $\tau_t^{\{o_i\}} \neq \tau_t^{\{o_j\}}$ may occur

3. Conflict Rules

R7 Conflict Detection: $d_P(p_i, p_j) > \epsilon \Rightarrow \text{Conflict}$

R8 Conflict Structural Rule: Conflict is a valid state element

R9 Coexist Rule: conflicting perceptions may coexist

R10 Priority Rule: higher weight perception selected

R11 Merge Rule: multiple perceptions merged

R12 Pending Rule: insufficient evidence $\Rightarrow$ suspend

4. History Rules

R13 History Append: $H_{t+1} = H_t \oplus e_t$

R14 History Influence: $G_{t+1} = \Phi_G(G_t, H_t, S_t)$

R15 Replay Rule: state can be approximated from history

5. Balance Rules

R16 Balance Evaluation: $B_t = \text{Bal}(S_t, O, H_t)$

R17 Rebalance: B_t outside safe region $\Rightarrow \Delta_{\text{bal}}$

R18 Dynamic Coherence: system coherence determined by balance

6. Dynamic Update

R19 World Flow: $W(S_t)$

R20 Full Update: $S_{t+1} = W(S_t) + \Delta_{\text{obs}} + \Delta_{\text{res}} + \Delta_{\text{goal}} + \Delta_{\text{bal}}$

R21 Observer Update: $\mathbf{o}_{t+1} = \Phi_O(\mathbf{o}_t, \mathbf{p}_t^{\mathbf{o}}, \mathbf{H}_t, \mathbf{B}_t)$